MARKING SCHEME

9.1 EXTERNALITIES

1985/AL/I/02 B	1992/AL/I/23 D	1998/AL/I/13 A	2006/AL/I/30 A (44%)	2013/DSE/I/20 D (69%)
1985/AL/I/23 D	1992/AL/I/24 C	1998/AL/I/19 A	2007/AL/I/30 B (63%)	2014/DSE/I/18 B (93%)
1986/AL/I/06 B	1993/AL/I/08 D	1999/AL/I/19 C	2008/AL/I/28 A (38%)	2014/DSE/I/19 B (81%)
1986/AL/I/25 D	1994/AL/I/08 B	1999/AL/I/27 B	2008/AL/I/30 C (45%)	2015/DSE/I/17 B (45%)
1988/AL/I/17 D	1994/AL/I/15 B	2000/AL/I/19 D	2009/AL/I/28 B (47%)	2018/DSE/I/23 B (73%)
1990/AL/I/25 A	1995/AL/I/23 D	2000/AL/I/20 B	2010/AL/I/28 D (73%)	2020/DSE/I/21 B
1990/AL/I/30 B	1996/AL/I/21 B	2003/AL/I/27 A	2012/AL/I/28 C (23%)	2022/DSE/I/22 B
1991/AL/I/01 C	1997/AL/I/27 C	2005/AL/I/29 C (37%)	SP/DSE/I/18 C	
1992/AL/I/22 D	1997/AL/I/28 B	2006/AL/I/29 A (71%)	2013/DSE/I/18 A (52%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1985/AL/I/7

(a) More of all goods, including housing, cars and education, is beneficial, but we cannot have more of all goods. If the government subsidizes education, more of it will be produced and consumed. Whether society benefits depends upon the value of the extra education (because of the subsidy) compared to the value of the other goods which the society would have to sacrifice in order to have the extra education (i.e. marginal benefit versus marginal cost). In general, because of the subsidy, the marginal cost of education to society would exceed its marginal benefit.

(b) Candidates are expected to know that this part of the question concerns the divergence between private and social benefits. If the educated cannot charge a fee for the benefit they generate to the uneducated, then from the society's point of view they will tend to under-invest in education, in the sense that the marginal cost of more education to them will fall short of the marginal social benefit. Subsidizing education by the government will therefore lead to marginal gain to society.

1986/AL/I/10

Level of Output	Marginal Revenue (MR)	Marginal Private Cost (MPC)	Marginal Damage	Marginal Social Cost (MSC)
100	180	130	34	164
200	180	140	32	172
300	180	150	30	180
400	180	160	28	188
500	180	170	26	196
600	180	180	24	204
700	180	200	22	222
800	180	310	20	330

(a) 600 units, where MR ($180) = MPC ($180) and the private gain is maximized ($150).

(b) 300 units, where social surplus is maximized ($24). If the factory owner wants to increase output to, say, 400 units, the marginal profit from this increment is $20, but the marginal damage to the neighbours is $28. The neighbours therefore will pay the factory owner $20 to cut output back to 300 units.

(c) Again 300 units, where the social surplus is maximum ($24). If the neighbours seek to reduce the factory's output to, say, 200 units, the marginal reduction in damage is $30. Since the marginal profit (from the 3rd 100 units) is $30, the factory owner will be able to pay and compensate neighbours to take the marginal damage.

1988/AL/I/7(a)
If congestion occurs, each user will slow down the speed of others, thereby imposing time cost upon one another without compensating them for this cost. The social time cost generated by one driver is the time cost of himself as well as the time cost he imposes upon all other drivers, whereas the private cost is the time cost to this one driver only.

A toll will reduce the traffic congestion, thereby reducing the likelihood of one user imposing time cost upon others.

2005/AL/I/7(a)
The presence of external effects, such as national pride or glory, may yield values to citizens at large, as most if not all Chinese in some measure enjoy a greater number of gold medals won in the Olympic Games. It would be highly difficult to collect payments for this type of external beneficial effects from all those enjoying the results of the training. (Transaction costs may be mentioned here.) Does national pride worth something to the citizens? Of course. But since payments are not enforced there is no way we can assess actually how much the citizens would be willing to pay.

2007/AL/I/7(a)
Social gains: Non-smokers' exposure to secondhand smoke is harmful to them health, and they are not compensated by the smokers who are responsible for such a negative externality. A ban on smoking can eliminate the existence of this negative externality.

Social costs: Smoking generates marginal benefit (MB) to smokers which exceeds its marginal social cost (MSC) at some levels. To maximize total social surplus (smokers' consumer surplus and tobacco companies' producer surplus), the 'efficient level of smoking' should be that smokers are allowed to smoke until their M(S)B equals MSC. A ban on smoking will lead to under-production and a deadweight loss.

Society will gain from a ban on smoking only if such social gains and larger than the social costs.

It is virtually impossible to show whether the social gains are greater or smaller than the social costs in this case, because there is no guideline to properly estimate these gains and costs. When people are not asked or not required to pay, as they do in the market, any such estimate is not reliable.

2008/AL/I/7(b)
There is a divergence between private and social costs in the sense that the anchor stores generate spillover benefits to the smaller shops.

There is no divergence between these spillovers are all captured by the owners of the shopping centers, and because the anchor stores are capturing their own beneficial spillovers by paying lower rents. On the other hand, the smaller shops, which pay higher rents, are actually buying the spillovers beneficial to them.

2015/DSE/I/12(d)
Negative externality / divergence between private and social costs. (1)
When the public transportation system is overcrowded (i.e., at full capacity), any additional passenger using the system would increase the time cost for other passengers without having to compensate them for this extra cost (implying a divergence between private and social costs). As marginal social cost is now higher than marginal social benefit, the number of passengers using the system would exceed its efficient level. (3)

2016/DSE/I/4
Taking photos may disturb other people in the restaurants/sharing the photos in the social media may disclose the design of dishes. Both possibilities may have an undesirable effect on the revenue/profit of the restaurants, but such effects are not compensated (financially or otherwise) by the photo-takers. Such photo-taking-and sharing behaviour may thus involve negative externality (external cost is involved). (3)

OR

Taking photos and sharing them in social media may have a promotion effect, attracting more customers-bringing higher revenue/profit-to the restaurant, but the photo takers do not receive compensation for such benefits. Such photo-taking-and sharing behaviour may thus involve positive externality (external benefit is involved). (3)

2017/DSE/I/4
(a) Positive externality. Restaurants in remote areas enjoy benefits from increased business due to the game developer's choice of hotspots locations without paying the developer. So external benefit exists and marginal social benefit is greater than marginal social cost. The number of existing hotspots near the restaurants is lower than that required by the efficiency level. (3)

(b) The restaurants may pay the game developer to increase the hotspots nearby. (2)
OR
The game developer may charge the restaurants for placing more hotspots near them. (2)

2019/DSE/II/4

Getting vaccinated protects oneself as well as lowering the chance of others being infected by flu. Since the person receiving vaccination is not compensated by others (whose chances of being infected are reduced), there exists external benefit, with marginal social benefit exceeding marginal social cost. This implies that the number of people getting vaccinated is below the efficient level. Subsidy can increase the quantity of vaccination towards the efficient level and thus narrow the gap between marginal social benefit and marginal social cost. **(4)**

2022/DSE/II/10a

(a) The broadcast would attract more customers and thus bring more business to the restaurants.Howeverin the absence of compensations from these restaurants (beneficiaries)tothe shopping malls (benefactors) for broadcasting the Olympic Gamesthere would exist a positive externality (or external benefit), with the marginal social benefit(MSB)from the broadcast exceeding the marginal private benefit (MPB). (3)

2022/DSE/II/10b

(2) (b) The difference between MSB and MPB implies under-provision of broadcast. A Market solution may involve the restaurants paying the malls to broadcast more games so that the numberofOlympic events broadcast would increase towards its efficient level. (2)

2024/DSE/II/2

(a) Negative externality may be involved when producers engaged in industrial activities which emit carbon dioxide do not have to pay compensation for the harm inflicted on others. (2)
As a result, equilibrium output determined by equality between marginal private (= social) benefit and marginal private (< social) cost would be higher than the socially efficient level (where marginal social benefit = marginal social cost). Such over-production resulting from the divergence between marginal private cost and marginal social cost implies economic inefficiency (i.e., inefficient allocation of the society's resources in producing such products). (2)

(b) Payment for carbon credits to engage in production activities that emit carbon dioxide would raise the producers' marginal private cost and thus lower equilibrium output towards the efficient level. At this lower level of output, the gap between marginal private cost and marginal social cost is reduced and economic efficiency is improved. (2)

2025/DSE/II/3

The tutorial teacher may benefit from an increase in income from advertisement on social media / from his tutorial classes due to an increase in popularity when people view his videos. (2)
If the increase in income equals the benefits enjoyed by the viewers of his videos at the margin, there would be no externality. (1)